

CHAPTER 3

Plant Kingdom

Algae

1. Read the following statements and choose the set of correct statements. (2024)

In the members of Phaeophyceae,

- A. Asexual reproduction occurs usually by biflagellate zoospores.
- B. Sexual reproduction is by oogamous method only.
- C. Stored food is in the form of carbohydrates which is either mannitol or laminarin.
- D. The major pigments found are chlorophyll a, c and carotenoids and xanthophyll.
- E. Vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.

Choose the correct answer from the options given below:

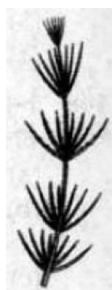
- a. A, C, D and E only b. A, B, C and E only
- c. A, B, C and C only d. B, C, D and E only

2. Which classes of algae possess pigment fucoxanthin and pigment phycoerythrin, respectively? (2023-Manipur)

- a. Phaeophyceae and Chlorophyceae
- b. Phaeophyceae and Rhodophyceae
- c. Chlorophyceae and Rhodophyceae
- d. Rhodophyceae and Phaeophyceae

3. Read the following statements and identify the characters related to the alga shown in the diagram (2022 Re)

- A. It is a member of Chlorophyceae
- B. Food is stored in the form of starch
- C. It is a monoecious plant showing oogonium and antheridium
- D. Food is stored in the form of laminarin or mannitol
- E. It shows dominance of pigments Chlorophyll a, c and Fucoxanthin



Choose the correct answer from the options given below:

- a. (C), (D) and (E) only b. (A) and (B) only
- c. (A), (B) and (C) only d. (A), (C) and (D) only

4. Which of the following is incorrectly matched? (2022)

- a. *Volvox* - Starch b. *Ectocarpus* - Fucoxanthin
- c. *Ulothrix* - Mannitol d. *Porphyra* - Floridean Starch

5. Hydrocolloid carrageen is obtained from: (2022)

- a. Phaeophyceae only
- b. Chlorophyceae and Phaeophyceae
- c. Phaeophyceae and Rhodophyceae
- d. Rhodophyceae only

6. Which of the following algae produce Carrageen? (2021)

- a. Brown algae b. Red algae
- c. Blue-green algae d. Green algae

7. Which of the following algae contains mannitol as reserve food material? (2021)

- a. *Gracilaria* b. *Volvox*
- c. *Ulothrix* d. *Ectocarpus*

8. Which of the following pairs is of unicellular algae? (2020)

- a. *Gelidium* and *Gracilaria* b. *Anabaena* and *Volvox*
- c. *Chlorella* and *Spirulina* d. *Laminaria* and *Sargassum*

9. Floridean starch has structure similar to: (2020)

- a. Amylopectin and glycogen
- b. Mannitol and algin
- c. Laminarin and cellulose
- d. Starch and cellulose

10. Phycoerythrin is the major pigment in: (2020-Covid)

- a. Blue green algae b. Green algae
- c. Brown algae d. Red algae

11. An example of colonial alga is (2017-Delhi)

- a. *Chlorella* b. *Volvox* c. *Ulothrix* d. *Spirogyra*

12. Which one of the following statements is wrong? (2016 - II)

- a. Agar-agar is obtained from *Gelidium* and *Gracilaria*.
- b. *Laminaria* and *Sargassum* are used as food.
- c. Algae increase the level of dissolved oxygen in the immediate environment.
- d. Algin is obtained from red algae, and carrageen from brown algae.

13. Male gametes are flagellated in: (2015)

- a. *Ectocarpus* b. *Spirogyra*
- c. *Polysiphonia* d. *Anabaena*

14. Isogamous condition with non-flagellated gametes is found in: (2013)

- a. *Fucus* b. *Chlamydomonas*
- c. *Spirogyra* d. *Volvox*

15. Select the wrong statement: (2013)
- Chlamydomonas* exhibits both isogamy and anisogamy and *Fucus* shows oogamy
 - Isogametes are similar in structure, function and behaviour
 - Anisogametes differ either in structure, function or behaviour
 - In oogamous reproduction, female gamete is smaller and motile, while male gamete is larger and non motile

Bryophytes

16. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R: (2023)
- Assertion A:** The first stage of gametophyte in the life cycle of moss is protonema stage.
- Reason R:** Protonema develops directly from spores produced in capsule.
- In the light of the above statements, choose the most appropriate answer from the options given below:
- A is not correct but R is correct.
 - Both A and R are correct and R is the correct explanation of A.
 - Both A and R are correct but R is NOT the correct explanation of A.
 - A is correct but R is not correct.
17. Gemmae are present in: (2021)
- Pteridophytes
 - Some Gymnosperms
 - Some Liverworts
 - Mosses
18. Which one is wrongly matched? (2018)
- Uniflagellate gametes - *Polysiphonia*
 - Biflagellate zoospores - Brown algae
 - Gemma cups - *Marchantia*
 - Unicellular organism - *Chlorella*

Pteridophytes

19. Identify the **incorrect** pair: (2024 Re)
- Sphenopsida - *Adiantum*
 - Pteropsida - *Dryopteris*
 - Psilopsida - *Psilotum*
 - Lycopsida - *Selaginella*
20. Identify the pair of heterosporous pteridophytes among the following: (2023)
- Equisetum* and *Salvinia*
 - Lycopodium* and *Selaginella*
 - Selaginella* and *Salvinia*
 - Psilotum* and *Salvinia*
21. Match List-I with List-II. (2023-Manipur)
- | List-I | | List-II | |
|--------|-------------|---------|--------------------|
| (A) | Pteropsida | (I) | <i>Psilotum</i> |
| (B) | Lycopsida | (II) | <i>Equisetum</i> |
| (C) | Psilopsida | (III) | <i>Adiantum</i> |
| (D) | Sphenopsida | (IV) | <i>Selaginella</i> |
- Choose the **correct** answer from the options given below:
- A-II, B-III, C-I, D-IV
 - A-III, B-I, C-IV, D-II
 - A-II, B-III, C-IV, D-I
 - A-III, B-IV, C-I, D-II
22. Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as: (2021)
- Heterosorus
 - Homosporous
 - Heterosporous
 - Homosorus

23. Strobili or cones are found in: (2020)
- Pteris*
 - Marchantia*
 - Equisetum*
 - Salvinia*
24. From evolutionary point of view, retention of the female gametophyte with developing young embryo on the parent sporophyte for some time, is first observed in (2019)
- Liverworts
 - Mosses
 - Pteridophytes
 - Gymnosperms
25. In bryophytes and pteridophytes, transport of male gametes requires: (2016 - I)
- Wind
 - Insects
 - Birds
 - Water

Gymnosperms

26. Which of the following is the **correct** match? (2024 Re)
- Gymnosperms : *Cedrus, Pinus, Sequoia*
 - Angiosperms : *Wolffia, Eucalyptus, Sequoia*
 - Bryophytes : *Polytrichum, Polysiphonia, Sphagnum*
 - Pteridophytes : *Equisetum, Ginkgo, Adiantum*
27. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R: (2023)
- Assertion A:** In gymnosperms the pollen grains are released from the microsporangium and carried by air currents.
- Reason R:** Air currents carry the pollen grains to the mouth of the archegonia where the male gametes are discharged and pollen tube is not formed.
- In the light of the above statements, choose the correct answer from the options given below:
- A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.
 - A is true but R is false.
28. Match List-I with List-II: (2022 Re)

List-I		List-II	
(A)	<i>Chlamydomonas</i>	(I)	Moss
(B)	<i>Cycas</i>	(II)	Pteridophyte
(C)	<i>Selaginella</i>	(III)	Alga
(D)	<i>Sphagnum</i>	(IV)	Gymnosperm

Choose the correct answer from the options given below.

- A-II, B-III, C-I, D-IV
 - A-III, B-I, C-II, D-IV
 - A-III, B-IV, C-II, D-I
 - A-III, B-II, C-I, D-IV
29. Which of the following statements is **incorrect** about gymnosperms? (2020-Covid)
- Male and female gametophytes are free living
 - Most of them have narrow leaves with thick cuticle
 - Their seeds are not covered
 - They are heterosporous

30. *Pinus* seed cannot germinate and established without fungal association. This is because : (2019)
 a. Its embryo is immature.
 b. It has obligate association with mycorrhizae.
 c. It has very hard seed coat.
 d. Its seeds contain inhibitors that prevent germination.
31. Which of the following statement is correct? (2018)
 a. Ovules are not enclosed by ovary wall in gymnosperms
 b. *Selaginella* is heterosporous, while *Salvinia* is homosporous
 c. Horsetails are gymnosperms
 d. Stems are usually unbranched in both *Cycas* and *Cedrus*
32. Winged pollen grains are present in (2018)
 a. Mustard b. *Cycas* c. Mango d. *Pinus*
33. Select the mismatch: (2017-Delhi)
 a. *Pinus* – Dioecious
 b. *Cycas* – Dioecious
 c. *Salvinia* – Heterosporous
 d. *Equisetum* – Homosporous
34. Conifers are adapted to tolerate extreme environmental conditions because of: (2016 - II)
 a. Thick cuticle b. Presence of vessels
 c. Broad hardy leaves d. Superficial stomata
35. Select the correct statement. (2016 - I)
 a. Gymnosperms are both homosporous and heterosporous
 b. *Salvinia*, *Ginkgo* and *Pinus* all are gymnosperms
 c. *Sequoia* is one of the tallest trees
 d. The leaves of gymnosperms are not well adapted to extremes of climate
36. Which one is wrong statement? (2015)
 a. *Mucor* has biflagellate zoospores
 b. Haploid endosperm is typical feature of gymnosperms
 c. Brown algae have chlorophyll *a* and *c* and fucoxanthin
 d. Archegonia are found in Bryophyta, Pteridophyta and Gymnosperms.
37. Read the following statements (A-E) and answer the question which follows them: (2013)
 A. In liverworts, mosses, and ferns gametophytes are free-living
 B. Gymnosperms and some ferns are heterosporous
 C. Sexual reproduction in *Fucus*, *Volvox* and *Albugo* is oogamous
 D. The sporophyte in liverworts is more elaborate than that in mosses
 E. Both *Pinus* and *Marchantia* are dioecious.

How many of the above statements are correct?

- a. Four b. One c. Two d. Three

38. Besides paddy fields, Cyanobacteria are also found inside vegetative part of: (2013)

- a. *Psilotum* b. *Pinus* c. *Cycas* d. *Equisetum*

Plant Life Cycles

39. Which of the following statements is true?

(2023-Manipur)

- a. All pteridophytes exhibit haplo-diplontic pattern.
 b. Seed bearing plants follow haplontic pattern
 c. Most algal genera are diplontic
 d. Most bryophytes do not have haplo-diplontic life cycle.

40. Match the plant with the kind of life cycle it exhibits:

List-I		List-II	
(A)	<i>Spirogyra</i>	(i)	Dominant diploid sporophyte vascular plant, with highly reduced male or female gametophyte
(B)	Fern	(ii)	Dominant haploid free-living gametophyte
(C)	<i>Funaria</i>	(iii)	Dominant diploid sporophyte alternating with reduced gametophyte called prothallus
(D)	<i>Cycas</i>	(iv)	Dominant haploid leafy gametophyte alternating with partially dependent multicellular sporophyte

Choose the correct answer from the options given below. (2022)

- a. A-ii B-iv C-i D-iii
 b. A-iv B-i C-ii D-iii
 c. A-ii B-iii C-iv D-i
 d. A-iii B-iv C-i D-ii

41. Life cycle of *Ectocarpus* and *Fucus* respectively are: (2017-Delhi)

- a. Haplontic, Diplontic
 b. Diplontic, Haplodiplontic
 c. Haplo-diplontic, Diplontic
 d. Haplo-diplontic, Haplontic

42. Zygotic meiosis is characteristic of: (2017-Delhi)

- a. *Marchantia* b. *Fucus*
 c. *Funaria* d. *Chlamydomonas*

Answer Key

1. (a) 2. (b) 3. (c) 4. (c) 5. (d) 6. (b) 7. (d) 8. (c) 9. (a) 10. (d)
 11. (b) 12. (d) 13. (a) 14. (c) 15. (d) 16. (b) 17. (c) 18. (a) 19. (a) 20. (c)
 21. (d) 22. (c) 23. (c) 24. (c) 25. (d) 26. (a) 27. (d) 28. (c) 29. (a) 30. (b)
 31. (a) 32. (d) 33. (a) 34. (a) 35. (c) 36. (a) 37. (d) 38. (c) 39. (a) 40. (c)
 41. (c) 42. (d)